

Your project should satisfy all the following criteria at minimum.

1. **Final Report Submission:**

- 1.1 The written Final Project Report is due on 07/29/2026 at 11:59 PM.
- 1.2 You can submit a single report out of a group. *You are also required to submit all the programs you created to complete the project as a separate Zip file.* For not submitting your Program(s), **50%** credit will be reduced from the entire weight of the project.
- 1.3 Late submission will NOT be accepted but for a reasonable and with documented late submission then 10% weight per hour will be reduced after the due date and time. Therefore, paper submitted 10 hours after the due date will simply deserve Zero credit.
- 1.4 The name of your submitted files should be in the structure of

FirstName_LastName_AdditionalInfo,

of the submitting student. Otherwise, **10%** of the credit will be reduced.
- 1.5 (2 points) There should be a clear and meaningful title of your project.
- 1.6 (3 points) The final submitted report should be no more than 10-pages including the cited references. But you can include an appendix with no page limitation.
- 1.7 (5 points) Technical conclusion of your completed project should be included in the final report.
- 1.8 (5 points) In your final report, a non technical report – called **Executive Sum-mary** – to be disseminated to the company executive board and/or the stakeholders should be included but not exceeding a single page in length.
- 1.9 (25 points) In the final report, at least the following items should be included: Explanatory data analysis (EDA) with visualization, feature selections, both regression and classification analysis, lasso and ridge via elastic net regression, GLMs with appropriate target distributions, tree-based analysis (included both correlated and uncorrelated trees, ensemble learning, etc.) dimension reduction, clustering, etc.
- 1.10 (5 points) The final report should be in a publication manuscript format, for ex-ample, IEEE Manuscript Templates for Conference Proceedings.
- 1.11 (5 points) All the key Data Mining concepts should be clearly formulated with proper details included. Partial credit will be given, but **only for work shown**. At most 10% of the total points will be given only for the final correct answer without the entire work.

- 1.12 Tables should be constructed using a word processor. **Do not simply copy–and–paste information from R and/or Python.**
- 1.13 Any graphics, if appropriate, (*should be included in your paper*) with their description should be together. Description on first page and graphics on seventh page, for example, will not be accepted.
- 1.14 Be sure to number your graphics and tables, give appropriate caption, i.e., title, label your axes which are properly readable.

2. **Presentation:**

- 2.1 The final presentation your course project is on July 29, 2026 from 6:00 PM to 8:50 PM.
- 2.2 The presentation time is from 15 to 30 minutes, depending upon the group size.
- 2.3 No limitations for the number of slides.
- 2.4 (5 points) Each member of the group should speak for about **Five Minutes** during the time given for the presentation.
- 2.5 (5 points) After the presentation of each group, there will be **Ten Minutes** of **Intermission Discussion**. You are required to participate in at least one Inter-mission Discussion.
- 2.6 (5 points) The slides should be clearly visual.
- 2.7 (5 points) Finished on given time with clear explanation.
- 2.8 (5 points) Your name on Zoom should reflect your official name which should match exactly whatever we see in WebCourses. Otherwise, we are not responsible to indentify you in the <https://www.ucf.edu/> system to assign your final letter grade of the course.
- 2.9 (5 points) During the entire presentation time of your group, you are required to turn on your video on Zoom, showing your face clearly.
- 2.10 (5 points) Clearly and concisely present the problem situation you have been trying to address in your project.
- 2.11 (5 points) Make your presentation effective covering all the major concepts/works you have done.
- 2.12 (5 points) Talk about your decision making recommendation to the executive committee.
- 2.13 (5 points) Talk about future directions which are left to be worked out in your project.